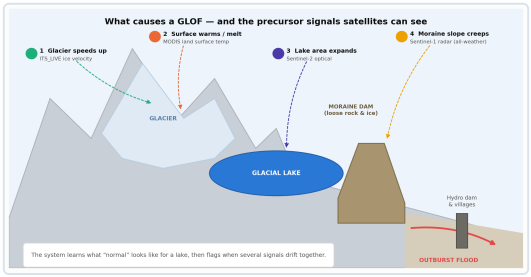
Satellite Watchdog for Himalayan Glacial Lake Floods

Major Project — 7th Semester

THE PROBLEM

High in the Himalaya, melting glaciers leave behind lakes held back only by **moraines** — natural walls of loose rock, mud and buried ice. Nobody built these walls, so they can give way. When one failed at South Lhonak Lake in Sikkim on 4 October 2023, the lake emptied in hours, destroyed the Teesta-III hydro dam at Chungthang and killed at least 40 people. The unsettling part: the warning signs were visible in satellite data beforehand, and researchers had even published about that slope two years earlier. The information existed — **no system was watching it**. Hundreds of similar lakes across India are still unwatched today.



Four warning signs that free satellites can already see — and the flood that follows if nobody is watching.

THE IDEA

Build a software system that keeps an eye on 15–25 glacial lakes automatically, using **free** satellite data from four independent sources: optical photos (Sentinel-2), all-weather radar that sees through monsoon cloud (Sentinel-1), glacier movement speed (NASA ITS_LIVE), and surface temperature (MODIS). Instead of trying to "predict a disaster" — impossible, since only one such event is properly recorded — the system learns what *normal* looks like for each lake over the years, and raises a flag when several signals start drifting away from normal at the same time. The output is a simple **ranked watch-list** on a dashboard for a State Disaster Management officer, with the supporting evidence and before/after images attached, so a human makes the final call.

NOVELTY — WHAT IS NEW HERE

- **Anticipatory, not forensic.** Existing work counts lakes or reconstructs floods *after* they happen; this watches for trouble *before*.
- **First multi-sensor precursor benchmark for the Indian Himalaya**, extending a Karakoram-only dataset to a new mountain region.
- **Radar added for the monsoon.** We measure how much observation all-weather SAR recovers when cloud blinds optical sensors — during the season of highest risk.
- **Learning without disasters.** A one-class formulation that works with only one confirmed event, where normal classification is statistically impossible.
- **Honest abstention.** If a lake can't be seen clearly the system says "cannot assess" instead of falsely reporting "safe".
- **Pre-validated gap.** The dataset's own authors list these directions as their future work.

TECHNOLOGIES USED

Layer	Tools
Language / compute	Python 3.11; Google Colab (free T4)
Geospatial cloud	Google Earth Engine; Copernicus Data Space
Raster & data cube	rasterio, GDAL, xarray, Dask
Vector GIS	geopandas, shapely
Machine learning	scikit-learn (Isolation Forest, One-Class SVM), LightGBM, PyTorch (LSTM / autoencoder)
Dashboard & reports	Streamlit, Folium, Plotly, ReportLab
Reproducibility	Git + GitHub, conda, MLflow

DATASETS NEEDED — ALL PUBLIC, TOTAL COST ZERO

Dataset	What it provides	Source
GLOFNet (2025)	Primary multimodal dataset: 600+ Sentinel-2 patches, 11.9 M ITS_LIVE velocity records, ~98,000 MODIS LST observations	arXiv 2510.10546 (mirror in Week 1)
Sentinel-2 L2A	10 m optical imagery — lake area, shoreline and surface change, ~5-day revisit	Copernicus / GEE
Sentinel-1 GRD (SAR)	All-weather day/night radar that sees through monsoon cloud — the key addition	Copernicus / GEE; ASF
ITS_LIVE	Global glacier surface velocity, 1980s to present — the acceleration signal	NASA MEaSUREs
MODIS MOD11A1	Daily 1 km land surface temperature — melt-season thermal index	NASA LP DAAC
Copernicus DEM 30 m	Elevation, slope, aspect and moraine geometry features	GEE
HMA GLOF Database	697 documented outburst floods (1833–2022) — label source for the retrospective test	Zenodo 7271187
ICIMOD inventories / RGI	Lake inventories and glacier outlines — used to select the study lakes	ICIMOD; GLIMS

Aligned with UN SDG 13.1 (climate resilience) · 11.5 (disaster losses) · 6.6 (mountain water ecosystems) | Decision support only — never an automated public alert.